AI Engineering
(Building Applications with Foundation Models)

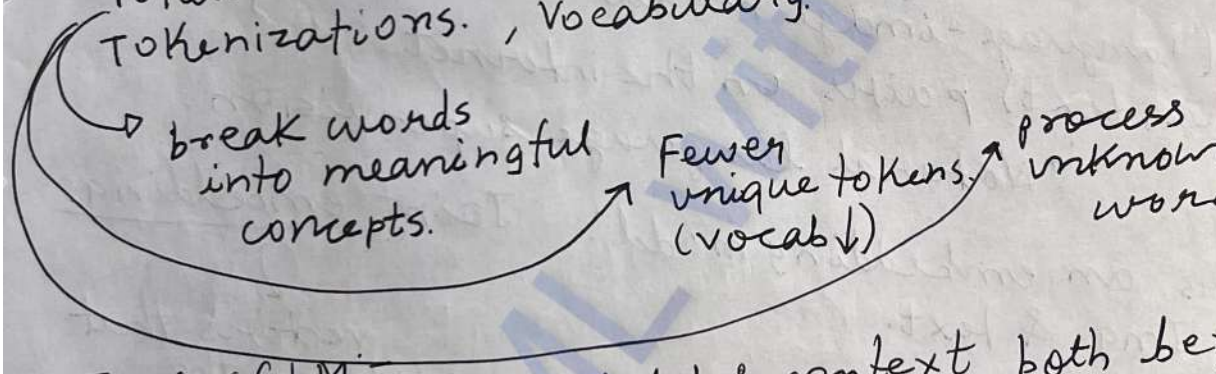
Preface

Build applications without writing code.

ch 1 :- Introduction to Building AI applications with Foundation Models

scale ↑, More powerful ↑
Model As a service. (No need to build models)
AI applications ↑ Entry barrier ↓
Build apps on top of foundation models.

Rise of AI Engg :-
Self-supervision
Language Model: Encodes statistical info about one/more language.
Token: Unit of LM.
Tokenizations., Vocabulary.



Types of LM:
Masked Language Model: context both before & after the missing tokens.
BERT.
Trained to predict missing tokens.

Autoregressive LM :- only preceding tokens as context

O/Ps of language model are open-ended (Gen AI).
Text completion task.
But needs post training to follow instructions.

Self-Supervision :- overcomes data-labelling bottleneck, allowing scaling.
Data labeling is expensive & time consuming

Aiml with Aayush

Model infers labels from data ↓ I/P

Eg: "I love street food"

Input (Context)	Output (next token)
<BOS>	I
<BOS>, I	love
⋮	⋮
⋮	<EOS>

Parameter: Updated through training process

From LLMs to Foundation Models
↳ (have multiple modalities)

Multiple modalities: Multimodal model
Next token cond^n on say both text & Img tokens.

CLIP (Language-image model)
(image, text) pairs on the internet.
↳ Natural language supervision.

CLIP is an embedding model: Joint embeddings
of Img & text. ↓
vectors that capture the meaning of data.

FMs are capable of wide range of tasks.

AI Engg techniques
↓ ↓ →
Prompt Engineering RAG Finetuning

FMs made it cheaper to develop AI apps and reduce time to market.

AI & ML with Ayush

From foundation Models to AI Engg :-

(i) General purpose AI capabilities.
(ii) Increased AI investments.
(iii) low entry barriers to building AI applications.
(Model As a Service)
Use Model APIs
Without them, using an AI model requires the infrastructure to host & serve the model.

Foundation Model Use Cases :-

Enterprise prefers applications with lowes risks.

Coding., Image & Videoproduction, Writing., Education, Conversational Bots, Information Aggregation, Data Organization, Workflow Automation.

Planning AI applications

→ Use Case Evaluation. → Why build ?
↳ competitors can use AI
↳ Miss Opportunity to boost profits & productivity
↳ Not left behind.
Human-in-the-loop (Human's in the AI decision making process).

→ Setting Expectations → Sucess : Business metric
terms definition.
Quality metrics
Latency metrics (TTFT, TPOT), Total latency
Token
cost metrics : per Inference Request
Interpretability, fairness

→ Milestone planning :-

→ Maintenance :- cost-benefit analysis

The AI Engineering Stack :-

- **Application development** — good, rigorous prompts.
- **Model development** — includes frameworks
- **Infrastructure** — model serving, managing data & compute, monitoring

AI Engg vs ML Engg :- (model dev.)

1) FMs use model instead of self-training.
2) Working with GPUs. (big).
3) Evaluation problem for open-ended o/Ps

More about adapting & evaluating models

- prompt-based
- Fine-tuning.

Model development

- Modeling & Training : Google's Tensorflow, HFs Transformers
  pretrain, finetune, post-train.
- Dataset Engg : open-ended FMs, so, data annotation quite challenging
  (Context retrieval, tokenization, deduplication qc.
- Inference optimization : Make models fast and cheap.

Application Development layer

- Evaluation
- Prompt engg
- AI interfaces.

AI & ML with Aayush

Evaluation:- mitigate risks
uncover new opportunities.

ML prblm: You have easy ground truth.
AI engg:- open-ended.

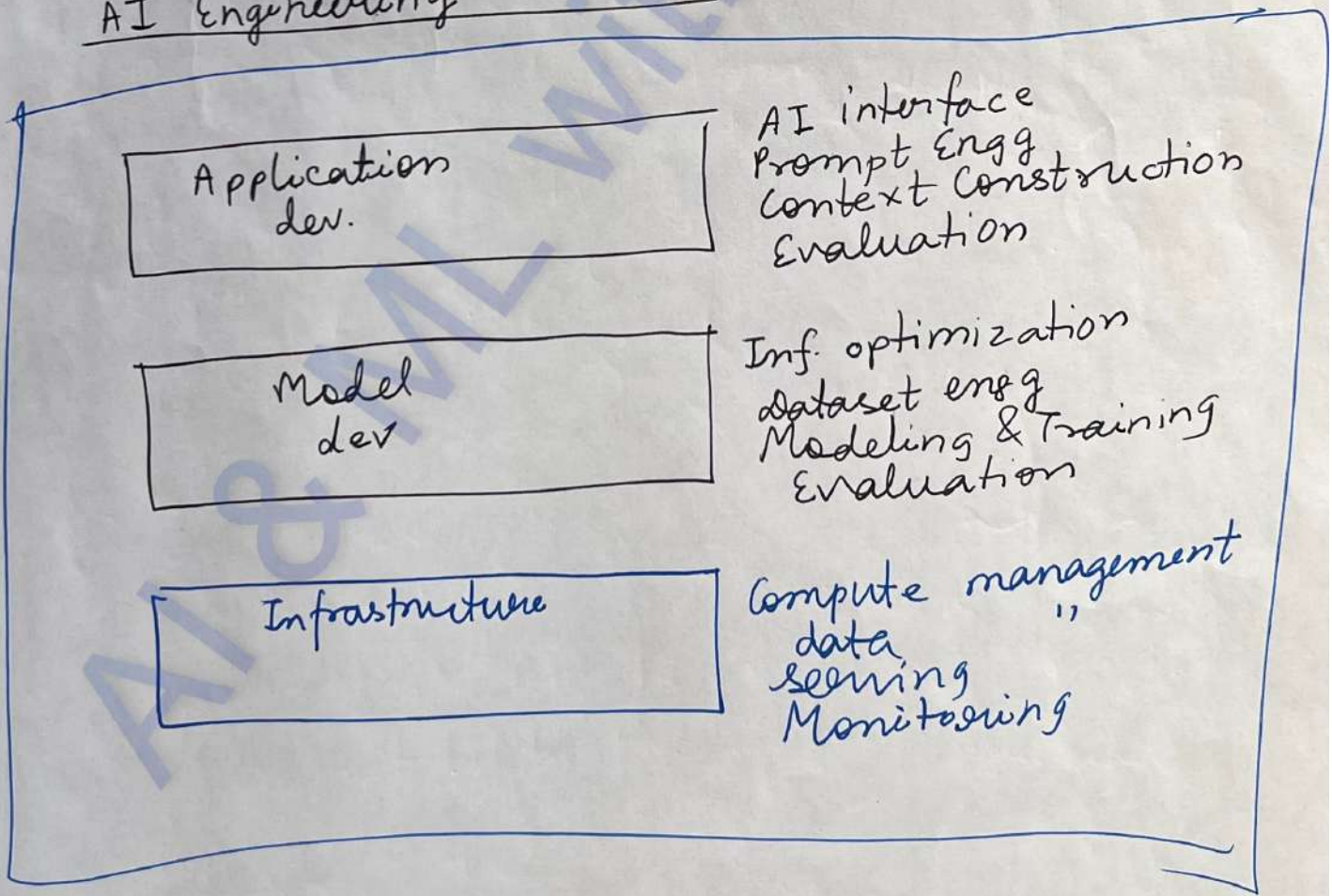
Prompt Engg & Context construction:

AI interface: users interact.

ML Engg: Data → Model → Product
AI Engg: Product → Data → Model
iterate

↓
(more involved in product building)

AI Engineering stack



AI & ML with Aayush